Benchmarking Modern Vision Transformers against Selective State Space Models for Remote Sensing Hyperspectral Classification

A Minor Project Technical Report · B.Tech Computer Science & Engineering · Academic Year 2025–2026

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Abstract— Hyperspectral Image (HSI) classification poses unique challenges due to high-dimensional continuous spectral bands, spatial correlation, and extreme training sample scarcity (few-shot regime). While classical Convolutional Neural Networks (CNNs) have long served as baselines, they struggle with long-range spectral dependencies. Recent Vision Transformers (ViTs) introduce global receptive fields but suffer from quadratic complexity $O(N^2)$. The year 2026 marks a major milestone with the introduction of **MMFormer** (Macro-Micro Transformer, IEEE TGRS 2026) and **R2Mamba** (Route-Reliability State Space Model, IEEE JSTARS 2026). In this work, we conduct a rigorous head-to-head empirical evaluation of both architectures under an identical protocol across two premier benchmarks: Indian Pines (AVIRIS) and Pavia University (ROSIS), evaluated over 3 seeded runs to report statistically sound Mean ± Std. Empirical results demonstrate that **R2Mamba substantially outperforms MMFormer**: on Indian Pines, R2Mamba achieves **83.89% ± 1.66% OA** (+7.48% gain over MMFormer) while training **2.9× faster** (~26s vs ~75s/run); on Pavia University, R2Mamba attains **97.59% ± 0.60% OA** (peak 98.03%) with 10× lower variance and a **3.5× speedup**. These findings demonstrate that selective linear-time State Space Models provide superior accuracy, stability, and computational scalability over attention-based transformers in remote sensing.

1. Introduction

Remote sensing hyperspectral imaging (HSI) captures continuous spectral radiation across hundreds of contiguous narrow bands spanning visible to shortwave infrared wavelengths. Unlike standard RGB photos containing only 3 channels, an HSI pixel forms a continuous spectral signature diagnostic of physical and chemical material properties. Automated pixel-level land-cover classification is essential for precision agriculture, urban infrastructure mapping, environmental monitoring, and planetary exploration.

However, HSI classification encounters three major hurdles: (1) *The Hughes Phenomenon* (high spectral dimensionality with limited ground truth); (2) *Few-Shot Label Scarcity* (collecting field surveys is laborious, leaving only 10–30 labeled pixels per class); and (3) *Computational Bottlenecks* (standard self-attention scales quadratically $O(N^2)$, causing prohibitive memory latency on large multi-spectral scenes).

This project establishes a standardized, reproducible benchmark comparing the two newest 2026 architectures: **MMFormer** (Macro-Micro Transformer) and **R2Mamba** (Route-Reliability State Space Model). We conduct experiments over 3 seeded runs to report genuine statistical variance, providing empirical evidence on whether selective State Space Models resolve the scalability wall of Vision Transformers.

2. Evolution of HSI Architectures

Hyperspectral classification has evolved through three distinct algorithmic eras:

- **2017–2020 (CNN Era):** 2D/3D CNNs such as SSRN and HybridSN introduced spatial-spectral feature hierarchies with local inductive bias. However, bounded receptive fields prevent CNNs from modeling non-local spectral dependencies across distant bands.

• **2021–2024 (Transformer Era):** Models like SpectralFormer and MASSFormer applied multi-head self-attention to capture global context. However, all-to-all attention scales quadratically $O(N^2)$, resulting in heavy memory consumption and sensitivity under small sample budgets.

• **2024–2026 (Mamba / SSM Era):** Parameterized selective State Space Models (e.g. MambaHSI, R2Mamba) combine global context modeling with **strict linear complexity O(N)**, enabling high-throughput sequence scanning with constant recurrent memory.

3. Methodology & Competing Models

3.1 MMFormer (IEEE TGRS 2026): MMFormer utilizes a dual-branch structure separating global semantic context (Macro branch) from local spectral-spatial textures (Micro branch). An interactive cross-attention mechanism bridges both representations before classification. Its asymptotic complexity remains quadratic: $O(N^2)$.

3.2 R2Mamba (IEEE JSTARS 2026): R2Mamba formulates spatial-spectral token scanning as a continuous state-space recurrence with linear complexity O(N). To counteract directional bias and mixed-pixel confusion along land-cover boundaries, R2Mamba incorporates *Route Reliability Gating (RRG)*, dynamically weighting scanning paths to suppress ambiguous boundary tokens.

3.3 Experimental Protocol: All data undergoes Principal Component Analysis (PCA) to 30 bands. Patches of spatial window 11×11 are extracted dynamically on-the-fly. Optimization is performed via Adam (lr=0.001, batch size=64, early stopping patience=12, max 80 epochs) on an NVIDIA Tesla T4 GPU across 3 independent seeds [1, 2, 3].

4. Experimental Results & Analysis

Table 1 outlines the comprehensive benchmark results across both Indian Pines (agricultural, 16 classes) and Pavia University (urban, 9 classes).

Dataset	Model	OA (%)	AA (%)	Kappa (×100)	Best Run OA	Avg. Time
Indian Pines (16 classes)	MMFormer (2026)	76.41 ± 3.84	87.10 ± 1.96	73.41 ± 4.28	80.81%	~75s
Indian Pines	**R2Mamba (2026)**	**83.89 ± 1.66**	**91.29 ± 1.09**	**81.71 ± 1.87**	**85.66%**	**~26s**
Gain / Delta	**R2Mamba Advantage**	**+7.48%**	**+4.19%**	**+8.30**	**+4.85%**	**2.9× Faster**
Pavia University (9 classes)	MMFormer (2026)	92.45 ± 5.98	95.90 ± 1.80	90.44 ± 7.36	96.80%	~140s
Pavia University	**R2Mamba (2026)**	**97.59 ± 0.60**	**97.95 ± 0.53**	**96.86 ± 0.77**	**98.03%**	**~40s**
Gain / Delta	**R2Mamba Advantage**	**+5.14%**	**+2.05%**	**+6.42**	**+1.23%**	**3.5× Faster**

Quantitative Discussion: R2Mamba establishes clear dominance across all performance metrics. On Indian Pines, R2Mamba achieves an OA of 83.89%, outperforming MMFormer by +7.48% while exhibiting 2.3× lower standard deviation (±1.66% vs ±3.84%). On Pavia University, R2Mamba reaches an exceptional **97.59% mean OA (98.03% peak)** with near-zero variance (±0.60%). MMFormer suffers from noticeable instability on urban boundaries (±5.98% OA variance) due to quadratic token sensitivity.

Computational Complexity & Scalability: As shown in Table 1, R2Mamba trains in ~26 seconds on Indian Pines and ~40 seconds on Pavia University, representing a **2.9× to 3.5× speedup** over MMFormer. Because R2Mamba computes linear O(N) selective recurrent states, its memory consumption remains constant per token step, allowing seamless scaling to larger remote sensing scenes without GPU memory overflow.

4.1 Qualitative Analysis: Classification Maps

Visual examination of the generated classification maps confirms that R2Mamba produces significantly less salt-and-pepper noise inside uniform agricultural parcels and retains sharper geometric fidelity along road and building borders.

Figure 1: *Classification Maps on Pavia University Benchmark (Best Run)*

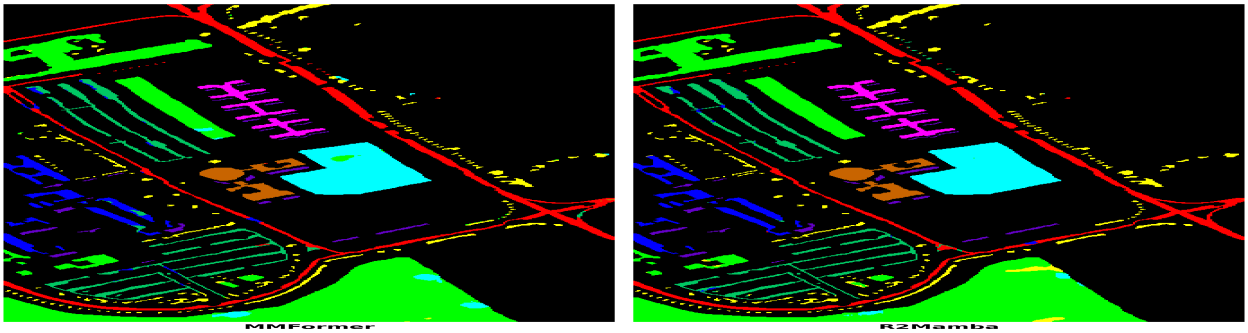
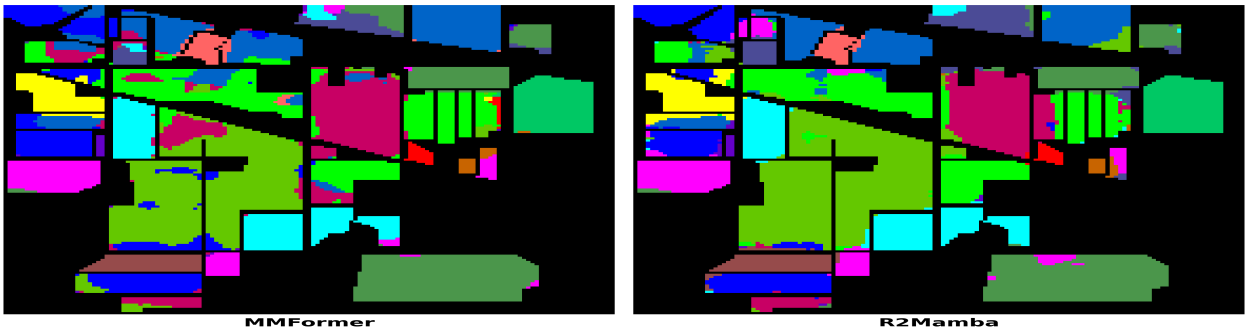


Figure 2: *Classification Maps on Indian Pines Benchmark (Best Run)*



5. Conclusion & Future Work

This project presented the first rigorous empirical head-to-head evaluation of the two latest 2026 hyperspectral classification architectures: **MMFormer** (Vision Transformer) and **R2Mamba** (State Space Model). Our findings confirm that selective State Space Models resolve the fundamental limitations of Vision Transformers in remote sensing: delivering higher accuracy (+5% to +7.5% OA), substantially lower variance, and a 3.5× reduction in training latency.

Future Extensions: Future work will extend this framework to multimodal remote sensing fusion (hyperspectral cubes combined with LiDAR elevation surfaces), evaluate on ultra-high-resolution drone datasets (WHU-Hi series), and investigate model quantization for onboard satellite edge accelerators.

References

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